

Exp 06 :

HTML :

```
<!DOCTYPE html>
<html>
<head>
  <title>Registration Form</title>
  <script src="validate.js"></script>
  <style>
    body {
      font-family: Arial;
      background-color: #f2f2f2;
    }
    .container {
      width: 420px;
      margin: auto;
      margin-top: 40px;
      background: white;
      padding: 20px;
      border-radius: 10px;
      box-shadow: 0px 0px 10px gray;
    }
    h2 {
      text-align: center;
    }
    label {
      font-weight: bold;
    }
    input, select {
      width: 100%;
      padding: 8px;
      margin-top: 5px;
      margin-bottom: 12px;
    }
    .radio-group {
      width: auto;
    }
    input[type="submit"] {
      background-color: green;
      color: white;
      border: none;
    }
  </style>
</head>
<body>
  <div class="container">
    <h2>Registration Form</h2>
    <div class="form">
      <div class="row">
        <div class="col">
          <label>First Name</label>
          <input type="text">
        </div>
        <div class="col">
          <label>Last Name</label>
          <input type="text">
        </div>
      </div>
      <div class="row">
        <div class="col">
          <label>Email</label>
          <input type="text">
        </div>
        <div class="col">
          <label>Phone</label>
          <input type="text">
        </div>
      </div>
      <div class="row">
        <div class="col">
          <label>Address</label>
          <input type="text">
        </div>
        <div class="col">
          <label>City</label>
          <input type="text">
        </div>
      </div>
      <div class="row">
        <div class="col">
          <label>State</label>
          <input type="text">
        </div>
        <div class="col">
          <label>Zip</label>
          <input type="text">
        </div>
      </div>
      <div class="row">
        <div class="col">
          <label>Gender</label>
          <input type="radio" value="Male"/> Male
          <input type="radio" value="Female"/> Female
        </div>
        <div class="col">
          <label>Age</label>
          <input type="text">
        </div>
      </div>
      <div class="row">
        <div class="col">
          <label>Password</label>
          <input type="password">
        </div>
        <div class="col">
          <label>Confirm Password</label>
          <input type="password">
        </div>
      </div>
      <div class="row">
        <div class="col">
          <label>Create Account</label>
          <input type="submit" value="Register">
        </div>
        <div class="col">
          <label>Already have an account?</label>
          <input type="button" value="Login">
        </div>
      </div>
    </div>
  </div>
</body>
</html>
```

```

input[type="submit"]:hover {
    background-color: darkgreen;
}
</style>
</head>
<body>
<div class="container">
    <h2>Registration Form</h2>
    <form name="myForm" onsubmit="return validateForm()">
        <label>Name:</label>
        <input type="text" name="name">
        <label>Email:</label>
        <input type="text" name="email">
        <label>Password:</label>
        <input type="password" name="password">
        <label>Phone:</label>
        <input type="text" name="phone">
        <label>Gender:</label>
        <input type="radio" name="gender" value="Male" class="radio-group"> Male
        <input type="radio" name="gender" value="Female" class="radio-group">
Female
        <br><br>
        <label>City:</label>
        <select name="city">
            <option value="">Select City</option>
            <option>Mumbai</option>
            <option>Pune</option>
            <option>Delhi</option>
        </select>
        <label>Age:</label>
        <input type="text" name="age">
        <input type="submit" value="Register">
    </form>
</div>
</body>
</html>

```

JS :

```
function validateForm() {
    var name = document.forms["myForm"]["name"].value;
    var email = document.forms["myForm"]["email"].value;
    var password = document.forms["myForm"]["password"].value;
    var phone = document.forms["myForm"]["phone"].value;
    var city = document.forms["myForm"]["city"].value;
    var age = document.forms["myForm"]["age"].value;
    var gender = document.querySelector('input[name="gender"]:checked');
    if (name == "" || email == "" || password == "" || phone == "" || city == ""
|| age == "" || gender == null) {
        alert("Please fill all fields");
        return false;
    }
    var emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (!email.match(emailPattern)) {
        alert("Invalid Email");
        return false;
    }
    var phonePattern = /^[0-9]{10}$/;
    if (!phone.match(phonePattern)) {
        alert("Phone must be 10 digits");
        return false;
    }
    if (password.length < 6) {
        alert("Password must be at least 6 characters");
        return false;
    }
    if (age < 1 || age > 100) {
        alert("Invalid Age");
        return false;
    }
    alert("Form Submitted Successfully");
    return true;
}
```

Exp 07 :

HTML :

```
<!DOCTYPE html>
<html>
<head>
  <title>Dynamic DOM Example</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <h1>Dynamic Content using DOM</h1>

  <!-- Buttons -->
  <button onclick="changeText()">Change Text</button>
  <button onclick="addElement()">Add Element</button>

  <!-- Content area -->
  <div id="content">Original Content</div>

  <!-- Link JavaScript -->
  <script src="script.js"></script>

</body>
</html>
```

CSS :

```
body {  
    font-family: Arial;  
    text-align: center;  
    margin-top: 50px;  
}
```

```
button {  
    padding: 10px 20px;  
    margin: 10px;  
    cursor: pointer;  
}
```

```
#content {  
    margin-top: 20px;  
    font-size: 18px;  
    color: blue;  
}
```

JS :

```
// Function to change text  
function changeText() {  
    document.getElementById("content").innerHTML = "Text changed dynamically!";  
}
```

```
// Function to add new paragraph  
function addElement() {  
    var newPara = document.createElement("p");  
    newPara.textContent = "New paragraph added using DOM!";  
  
    document.body.appendChild(newPara);  
}
```

Exp 08 :

HTML :

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple Calculator</title>
  <link rel="stylesheet" href="style.css">
  <script src="script.js"></script>
</head>
<body>
  <div class="container">
    <h2>Simple Calculator</h2>
    <label>Number 1:</label>
    <input type="text" id="n1">
    <label>Number 2:</label>
    <input type="text" id="n2">
    <label>Operator:</label>
    <select id="operator">
      <option value="+">Addition (+)</option>
      <option value="-">Subtraction (-)</option>
      <option value="*">Multiplication (*)</option>
      <option value="/">Division (/)</option>
    </select>
    <!--
    ===== Button Based Calculator (Alternative Method) =====
    <div class="buttons">
      <button onclick="add()">Add</button>
      <button onclick="subtract()">Subtract</button>
      <button onclick="multiply()">Multiply</button>
      <button onclick="divide()">Divide</button>
    </div>
    -->
    <button onclick="calculate()">Calculate</button>
    <label>Result:</label>
    <input type="text" id="result" readonly>
  </div>
</body>
</html>
```

CSS :

```
body {
    font-family: Arial;
    background-color: #f2f2f2;
}
.container {
    width: 300px;
    margin: auto;
    margin-top: 80px;
    background-color: white;
    padding: 20px;
    border-radius: 10px;
    box-shadow: 0px 0px 10px gray;
}
h2 {
    text-align: center;
}
label {
    font-weight: bold;
}
input, select {
    width: 100%;
    padding: 8px;
    margin: 8px 0;
}
button {
    width: 100%;
    padding: 10px;
    background-color: green;
    color: white;
    border: none;
    cursor: pointer;
}
button:hover {
    background-color: darkgreen;
}
```

JS :

```
function calculate() {
    var num1 = parseFloat(document.getElementById("n1").value);
    var num2 = parseFloat(document.getElementById("n2").value);
    var op = document.getElementById("operator").value;
    var result;
    switch(op) {
        case "+":
            result = add(num1, num2);
            break;
        case "-":
            result = subtract(num1, num2);
            break;
        case "*":
            result = multiply(num1, num2);
            break;
        case "/":
            result = divide(num1, num2);
            break;
        default:
            result = "Invalid";
    }
    document.getElementById("result").value = result;
}

function add(a, b) {
    return a + b;
}

function subtract(a, b) {
    return a - b;
}

function multiply(a, b) {
    return a * b;
}

function divide(a, b) {
    if (b == 0) {
        return "Cannot divide by zero";
    }
    return a / b;
}
```